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BATCH 15

1. Algorithm

Step 1: Start.

Step 2: Declare integer variables `m`, `n`, `choice`, `i`, and `j`.

Step 3: Input two integers `m` and `n`.

Step 4: Validate the input:

- Both `m` and `n` must be positive.
- `m` must be less than or equal to `n`.
- If the input is invalid, display an error message and input the values again.

Step 5: Display the menu:

1. Multiplication tables of `m` and `n`
2. Multiplication grid
3. Consecutive integers
4. Exit

Step 6: Input `choice`.

Step 7: Use a `switch` statement:

Case 1 – Multiplication Tables

1. Display the heading.
2. Set `i = 1`.
3. Repeat while `i <= 10`:
 - Calculate `m * i`.
 - Calculate `n * i`.
 - Display both multiplication-table entries side by side.
 - Increment `i`.

Case 2 – Multiplication Grid

1. Display column headings from 1 to `n`.
2. For each `i` from 1 to `m`:

- Display the row number i .
- For each j from 1 to n , display $i * j$.

Case 3 – Consecutive Integers

- For each i from m to n :
 - Display i .
 - Using another loop, display $i, i+1, i+2, i+3, i+4, i+5$.

Case 4 – Exit

- Display an exit message.

Default

- Display an error message indicating that the menu choice is invalid.

Step 8: Repeat Steps 5–7 until `choice` is 4.

Step 9: Stop.

2. Test-Case Table

Test Case	Input	Operation	Expected Output	Actual Output	Result
1	<code>m=3, n=5, Choice 1</code>	Multiplication tables	Tables of 3 and 5 from $\times 1$ to $\times 10$ displayed side by side	Tables correctly displayed	Pass
2	<code>m=3, n=5, Choice 2</code>	Multiplication grid	3 rows $\times$ 5 columns containing products	Correct 3×5 grid displayed	Pass
3	<code>m=3, n=5, Choice 3</code>	Consecutive integers	3: 3 4 5 6 7 8, through 5: 5 6 7 8 9 10	Correct sequences displayed	Pass
4	<code>m=1, n=1, Choice 1</code>	Boundary input	Table of 1 displayed in both columns	Correctly displayed	Pass
5	<code>m=0, n=5</code>	Invalid <code>m</code>	Error message and request input again	Error displayed	Pass
6	<code>m=3, n=0</code>	Invalid <code>n</code>	Error message and request input again	Error displayed	Pass
7	<code>m=5, n=3</code>	<code>m > n</code>	Error message and request input again	Error displayed	Pass
8	<code>m=3, n=5, Choice 9</code>	Invalid menu choice	Invalid menu choice. Please enter 1 to 4.	Correct error displayed	Pass
9	<code>m=3, n=5, Choice 4</code>	Exit	Program terminates	Program terminated	Pass

3. C Program

```
#include <stdio.h>
```

```
int main(void)
{
```

```

int m, n, choice, i, j;

/* Input m and n */
printf("Enter m and n (m <= n): ");
scanf("%d%d", &m, &n);

/* Validate m and n */
while (m <= 0 || n <= 0 || m > n)
{
    printf("Invalid input. Enter positive integers with m
<= n: ");
    scanf("%d%d", &m, &n);
}

do
{
    /* Display menu */
    printf("\n1. Multiplication tables of m and n\n");
    printf("2. Multiplication grid\n");
    printf("3. Consecutive integers\n");
    printf("4. Exit\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            /* Multiplication tables */
            printf("\nMultiplication Tables of %d and %d
"
                "(m=%d, n=%d)\n",
                m, n, m, n);

            for (i = 1; i <= 10; i++)
            {
                printf("%d x %d = %-10d"
                    "%d x %d = %d\n",
                    m, i, m * i,
                    n, i, n * i);
            }
            break;

        case 2:
            /* Multiplication grid */

```

```

printf("\nMultiplication Grid (m=%d, n=%d)
\n\n",
        m, n);

/* Column headings */
printf(" ");
for (j = 1; j <= n; j++)
{
    printf("%4d", j);
}

printf("\n ");

for (j = 1; j <= n; j++)
{
    printf("----");
}

printf("\n");

/* Grid */
for (i = 1; i <= m; i++)
{
    printf("%2d |", i);

    for (j = 1; j <= n; j++)
    {
        printf("%4d", i * j);
    }

    printf("\n");
}
break;

case 3:
/* Consecutive integers */
printf("\nConsecutive Integers from %d to %d
"
        "(m=%d, n=%d)\n",
        m, n, m, n);

for (i = m; i <= n; i++)
{
    printf("%d:", i);

```

```

        for (j = 0; j <= 5; j++)
        {
            printf(" %d", i + j);
        }

        printf("\n");
    }
    break;

case 4:
    printf("\nExiting program.\n");
    break;

default:
    printf("\nInvalid menu choice. "
        "Please enter 1 to 4.\n");
}

} while (choice != 4);

return 0;
}

```

5. Sample Execution

For:

```

m = 3
n = 5
Choice 1

```

Choice 2

```
dhruv@Dhruvs-MacBook-Air ~ % ./a.out
```

```
Enter m and n (m <= n): 3 5
```

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit

```
Enter your choice: 1
```

```
Multiplication Tables of 3 and 5 (m=3, n=5)
```

3 x 1 = 3	5 x 1 = 5
3 x 2 = 6	5 x 2 = 10
3 x 3 = 9	5 x 3 = 15
3 x 4 = 12	5 x 4 = 20
3 x 5 = 15	5 x 5 = 25
3 x 6 = 18	5 x 6 = 30
3 x 7 = 21	5 x 7 = 35
3 x 8 = 24	5 x 8 = 40
3 x 9 = 27	5 x 9 = 45
3 x 10 = 30	5 x 10 = 50

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit

```
Enter your choice: 2
```

```
Multiplication Grid (m=3, n=5)
```

	1	2	3	4	5
1	1	2	3	4	5
2	2	4	6	8	10
3	3	6	9	12	15

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit

```
Enter your choice:
```

Choice 3

```
Enter your choice: 3

Consecutive Integers from 3 to 5 (m=3, n=5)
3: 3 4 5 6 7 8
4: 4 5 6 7 8 9
5: 5 6 7 8 9 10

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
```

Invalid choice

```
Enter your choice: 9

Invalid menu choice. Please enter 1 to 4.

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
```

Choice 4

```
Enter your choice: 4

Exiting program.
dhruv@Dhruvs-MacBook-Air ~ %
```

testing result: The program compiled successfully and produced the expected results for the tested valid, invalid, and boundary cases.